

The Bohr-Rutherford Model

Chemical properties of elements are related to the energy changes that take place when their atoms gain, lose, or share electrons to obtain a filled valence shell.

stable octet

The farther the valence electron is from the positive nucleus, the more easily it is removed, and the more reactive the atom is.

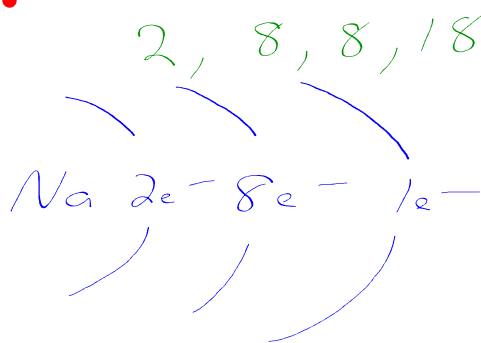
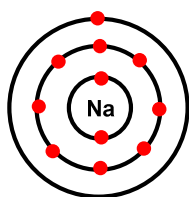
The outermost shell is known as the **valence shell**

Electrons in valence shell are called **valence electrons**

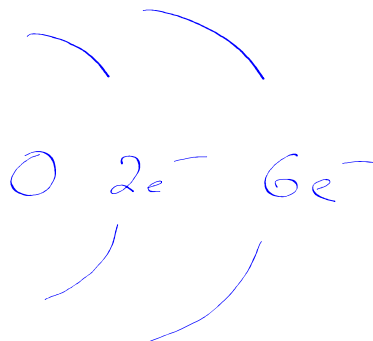
Bohr Diagrams used to show the location of all electrons in an atom

Electrons •

Sodium

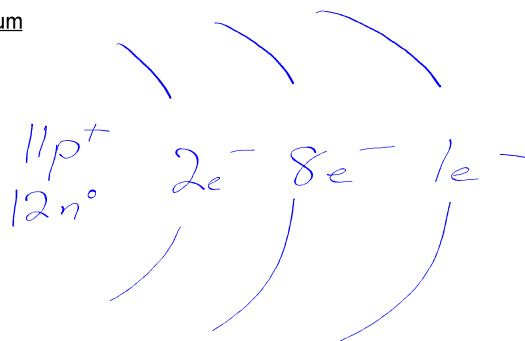


Oxygen

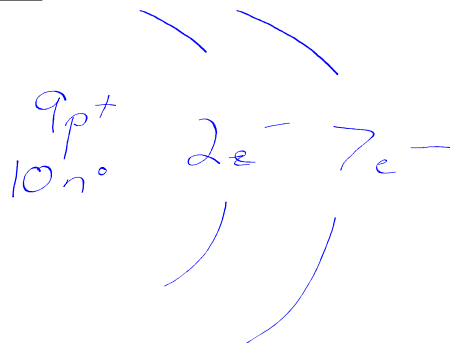


Bohr-Rutherford Diagrams also show the details of the nucleus

Sodium



Fluorine



Lewis Dot Diagrams shows the number of *valence* electrons only

- Add electrons starting from the top (or north) of the symbol, moving clockwise
- Once 4 electrons are added, start making pairs of electrons

Sodium



Nitrogen



Sulfur



Magnesium ion



Chloride



Argon



Lithium



Helium

